

## 4.4 Indeterminate Forms and L'Hôpital's Rule - continued

### RECALL: L'Hôpital's Rule

Suppose that  $f(a) = g(a) = \underline{\hspace{2cm}}$ , that  $f$  and  $g$  are  $\underline{\hspace{2cm}}$  on an open interval  $I$  containing  $a$ , and that  $\underline{\hspace{2cm}}$  on  $I$  if  $x \neq a$ . If

$$\begin{array}{l} \lim_{x \rightarrow a} f(x) = 0 \\ \lim_{x \rightarrow a} g(x) = 0 \end{array} \quad \text{or} \quad \begin{array}{l} \lim_{x \rightarrow a} f(x) = \pm\infty \\ \lim_{x \rightarrow a} g(x) = \pm\infty \end{array}$$

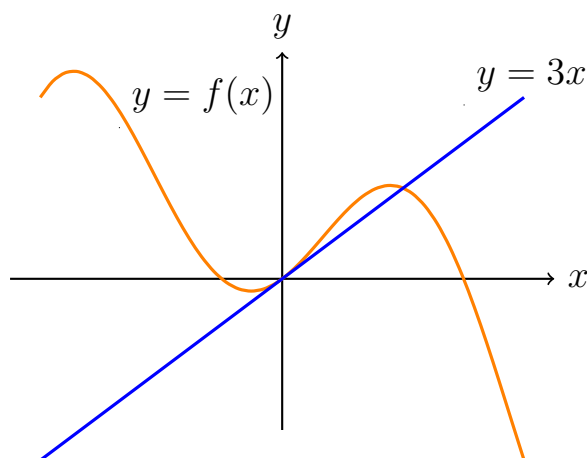
then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} =$$

assuming that the limit on the right side of this equation exists.

### Example 3

The graph of a function  $f$  and its tangent line at  $x = 0$  are shown below:



What is the value of  $\lim_{x \rightarrow 0} \frac{f(x)}{e^x - 1}$ ?

### Indeterminate Forms of the Type $0 \cdot \infty$

If we have  $\lim_{x \rightarrow a} f(x) \cdot g(x) = 0 \cdot \infty$ , we rewrite the limit as:

$$\lim_{x \rightarrow a} f(x) \cdot g(x) =$$

Apply L'Hôpital's Rule when appropriate.

### Example 4

Evaluate  $\lim_{x \rightarrow 0^+} (\sin x) \ln x$ .

### Indeterminate Forms of the Type $\infty - \infty$

If we have  $\lim_{x \rightarrow a} (f(x) - g(x)) = \infty - \infty$ , rewrite the limit as a

\_\_\_\_\_.

Apply L'Hôpital's Rule when appropriate.

### Example 5

Evaluate  $\lim_{x \rightarrow \infty} (\sqrt{x} - \sqrt{x+1})$

### Indeterminate Forms of the Type $1^\infty$ , $\infty^0$ or $0^0$

If  $\lim_{x \rightarrow a} f(x)^{g(x)}$  leads to an indeterminate power form, we transform the limit using the \_\_\_\_\_ function:

$$\ln \left( \lim_{x \rightarrow a} f(x)^{g(x)} \right) =$$

by the continuity of limits.

If the latter limit converges (possibly evaluated with L'Hôpital's Rule), then:

$$\lim_{x \rightarrow a} g(x) \cdot \ln(f(x)) = \quad \Rightarrow \quad \lim_{x \rightarrow a} f(x)^{g(x)} = \quad .$$

### Example 6

Evaluate  $\lim_{x \rightarrow 0} (\cos(x))^{1/x^2}$

**Extra Examples:**

1. Evaluate  $\lim_{x \rightarrow 0} \left( \frac{1}{x} - \frac{1}{\sin x} \right)$

2. Evaluate  $\lim_{x \rightarrow 0^+} x^x$